

CONTACT INFORMATION	Sabanci Universitesi, FASS 2029, Orta Mahalle, Tuzla, Istanbul, Turkiye	E-mail: caglar.tunc@sabanciuniv.edu Personal Website: www.caglartunc.com
RESEARCH INTERESTS	6G, O-RAN, V-RAN, NTN, and Digital Twins. Stochastic design and analysis of wireless networks. Applications of machine learning for Beyond 5G.	
EDUCATION	NYU Tandon School of Engineering , New York, NY, USA Ph.D., Electrical and Computer Engineering, September 2016 - February 2022 ▪ CGPA: 3.97/4.00 ▪ Advisor: Prof. Dr. Shivendra Panwar ▪ Dissertation title: "Mitigating the Challenges of Low-Latency Services in Future Wireless Networks" ▪ Ernst Weber Fellowship, Ph.D. ▪ Related Courses Taken: Wireless Communications, Massive MIMO, Advanced Machine Learning, Convex/Nonsmooth Optimization, Information Theory, Advanced Signal Processing, Queuing Theory Bilkent University , Ankara, Turkey M.S., Electrical and Electronics Engineering, September 2013 - June 2016 ▪ CGPA: 3.84/4.00 ▪ Advisor: Prof. Dr. Nail Akar ▪ Thesis title: "Energy Management in Energy Harvesting Wireless Sensor Nodes With Lifetime Constraints" ▪ TUBITAK Graduate Scholarship B.S., Electrical and Electronics Engineering, August 2009 - June 2013 ▪ CGPA: 3.77/4.00 ▪ Comprehensive Scholarship by Placement Examination (OYS) Ankara Ataturk Anatolian High School , Ankara, Turkey September 2005 - June 2009	
WORK EXPERIENCE	Sabancı University , İstanbul, Türkiye <i>Assistant Professor of Electronics Engineering</i> September 2024 - Present Turkcell 6GEN Lab , Maltepe, İstanbul, Türkiye <i>6G Researcher</i> February 2024 - September 2024 ▪ Investigating technologies and verticals for 6G networks ▪ Coordinating national and EU projects AT&T Labs Research , Bedminster, NJ, USA <i>Senior Inventive Scientist</i> March 2022 - January 2024 ▪ Research and development of O-RAN-based networking ▪ Development of the Digital Twin platform for cellular network automation and testing Samsung Research America , Berkeley Heights, NJ, USA <i>Intern, Standard and Mobility Innovation (SMI)</i> May 2020 - August 2020 ▪ Electric and magnetic fields (EMF)-based power control for Massive MIMO Futurewei Technologies , Bridgewater, NJ, USA <i>Systems Engineering Intern</i> May 2019 - August 2019 ▪ Data/statistical analysis and machine learning-based prediction/performance improvement of link adaptation in 5G-NR <i>Systems Engineering Intern</i> May 2018 - August 2018 ▪ Optimizing link adaptation in 5G-NR using machine learning techniques <i>Systems Engineering Intern</i> May 2017 - August 2017 ▪ Carrier aggregation in LTE-Advanced ASELSAN , Ankara, Turkey <i>Part-Time System Engineer</i> December 2012 - June 2013 ▪ Design of Wireless Communication Devices and Networks for Police and Gendarmerie	

ACADEMIC EXPERIENCE	NYU Wireless, NYU Tandon School of Engineering, New York, NY, USA	
	<i>Research Assistant</i>	
	<i>Teaching Assistant</i>	September 2016 - February 2022
	▪ Internet Architecture and Protocols, Graduate Core Course ▪ Fundamentals of Communication Theory, Undergraduate Core Course ▪ Communication Networks, Undergraduate Core Course ▪ Probability and Stochastic Processes, Graduate Core Course	
	Bilkent University, Ankara, Turkey	Fall 2013 - Spring 2016
	<i>Research Assistant</i>	
	<i>Graduate Teaching Assistant</i>	
	▪ Internet Architecture and Protocols, EEE536, Graduate Elective Course ▪ Telecommunications I, EEE431, 4th year Undergraduate Elective Course ▪ Microprocessors, EEE212, 2nd year Undergraduate Course ▪ Project & Laboratory Coordinator Assistant	
PROFESSIONAL SERVICE	Associate Editor	
	<i>IEEE Wireless Communications Letters</i>	March 2024 - Present
	Technical Program Committee Member	
	<i>IEEE Conference on Standards for Communications and Networking (CSCN 2024)</i>	
	<i>Track on Softwarization, Slicing, Automation and Network Management</i>	
BOOK CHAPTER	▪ K. Joshi, TX. Tran and C. Tunc. "Digital Twin for Beyond 5G", <i>AI in Wireless for Beyond 5G Networks</i>, IEEE Press Series on Digital and Mobile Communications, CRC Press, February 2024.	
JOURNAL PUBLICATIONS	▪ C. Tunc and S. Panwar. "Mitigating the Impact of Blockages in Millimeter-Wave Vehicular Networks through Vehicular Relays", <i>IEEE Open Journal of Intelligent Transportation Systems</i>, July 2021. ▪ C. Tunc, MF. Özkoç, F. Fund and S. Panwar. "The Blind Side: Latency Challenges in Millimeter Wave Networks for Connected Vehicle Applications", <i>IEEE Transactions on Vehicular Technology</i>, December 2020. ▪ E. O. Gamgam, C. Tunc and N. Akar. "On the Queuing Model of the Energy-Delay Trade-Off in Wireless Links with Power Control and Link Adaptation", <i>IEEE Transactions on Communications</i>, February 2019. ▪ N. Akar, C. Tunc, M. A. Gaertner and F. Erden. "Performance of Shortest Cumulative Access Time First (SCATF) Disk Scheduling Algorithms", <i>The Turkish Journal of Electrical Engineering & Computer Sciences</i>, July 2017. ▪ C. Tunc and N. Akar. "Markov Fluid Queue Model of an Energy Harvesting IoT Device with Adaptive Sensing", <i>Performance Evaluation</i>, May 2017. ▪ C. Tunc and N. Akar. "Fixed-point Analysis of a Network of Routers with Persistent UDP and TCP Flows and Class-based Weighted Fair Queuing" <i>Telecommunication Systems</i>, July 2016. ▪ C. Tunc and N. Akar. "Mapping Time-varying IP Traffic to Flexible Optical Paths in Flexgrid Optical Networks" <i>Photonic Network Communications</i>, August 2014.	
WORKING PAPERS	▪ K. Czapiga, S. Isci, Y. Karon, V. Kounev, TX Tran and C. Tunc. "AI-Accelerated Digital Twins for 6G", in progress.	
PATENTS	▪ Z. Lin, S. Das, C. Tunc and J. Zhang. "Apparatus and method for managing the exposure to electric and magnetic fields (EMF)", US11064443B1, July 2021.	
CONFERENCE PUBLICATIONS	▪ C. Tunc. "Optimal Digital Twinning of Random Systems with Twinning Rate Constraints", accepted to <i>IEEE International Workshop on Computer Aided Modeling and Design of Communication Links and Networks (CAMAD 2024)</i>, <i>1st Workshop on Systems and Technologies for IT/OT Integration in industry 4/5.0 Environments (SITE 2024)</i>.	

	▪ MF. Özkoç, C. Tunc and S. Panwar. "Data-Driven Beamforming Codebook Design to Improve Coverage in Millimeter Wave Networks", <i>IEEE 95th Vehicular Technology Conference: VTC2022-Spring</i>. ▪ C. Tunc and S. Panwar. "Analysis of Outage Probability and Duration in Millimeter Wave Vehicle-to-Infrastructure Networks", <i>IEEE 92nd Vehicular Technology Conference: VTC2020-Fall</i>, Victoria, BC Canada. ▪ C. Tunc and S. Panwar. "Optimal Transmission Policies for Energy Harvesting Age of Information Systems with Battery Recovery", <i>2019 Asilomar Conference on Signals, Systems, and Computers</i>, Pacific Grove, CA, USA. ▪ C. Tunc, MF. Özkoç and S. Panwar. "Millimeter Wave Coverage and Blockage Duration Analysis for Vehicular Communications", <i>IEEE 90th Vehicular Technology Conference: VTC2019-Fall</i>, Honolulu, Hawaii, USA. ▪ C. Tunc and N. Akar. "Efficient Transport of Time-varying IP Traffic in Flexi-grid Optical Networks", <i>Signal Processing and Communications Applications Conference (SIU)</i>, Trabzon, Turkey, April 2014.
PRESENTATIONS	▪ C. Tunc K. Duran, B. Bilgin, G. Kalem, B. Canberk. "DTRAN: A Special Use Case of RAN Optimization using Digital Twin", <i>European Conference on Networks and Communications (EuCNC 2024) & 6G Summit</i>, Antwerp, Belgium, June 2024. ▪ C. Tunc and N. Akar. "Performance Modeling of Delay-based Dynamic Speed Scaling", <i>The Ninth International Conference on Matrix-Analytic Methods in Stochastic Models (MAM9)</i>, Budapest, Hungary, June 2016.
DISSERTATION	C. Tunc . "Mitigating the Challenges of Low-Latency Services in Future Wireless Networks." <i>Ph.D. Dissertation</i> , New York University, May 2022.
THESIS	C. Tunc . "Energy Management in Energy Harvesting Wireless Sensor Nodes with Lifetime Constraints." <i>MS Thesis</i> , Bilkent University, June 2016.
TEACHING	▪ EE311 - Introduction to Signal Processing and Information Systems ▪ EE312 - Discrete Time Signals & Systems
LANGUAGES	English: Fluent, Spanish: Moderate, German: Beginner, Turkish: Native
COMP. SKILLS	Python, MATLAB/CVX/Gurobi, Torch, TensorFlow, VHDL, Java, R, AMPL, Assembly.
AWARDS & ACHIEVEMENTS	▪ Ernst Weber Fellowship (Ph.D.), NYU Tandon School of Engineering ▪ TUBITAK Graduate Scholarship ▪ Bilkent University Master of Science Study Full Scholarship ▪ Bilkent University High Honor Student (2009-2010 Fall, 2009-2010 Spring, 2010-2011 Fall, 2010-2011 Spring, 2011-2012 Fall, 2012-2013 Fall, 2012-2013 Spring) ▪ Bilkent University Honor Student (2011-2012 Spring) ▪ TUSIAD's 'Bu Gençlikte İş Var' Honourable Mention Award ▪ Ranked 420th in University Entrance Exam among 1.4 million candidates